

CORE PURE / FURTHER CALCULUS

Integration by Substitution

Worked solutions

15 questions · 113 marks

Questions

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1. Show that

$$\int_0^{\pi/2} \frac{\cos^5 \theta}{2 + \sin \theta} d\theta = 9 \ln\left(\frac{3}{2}\right) - \frac{41}{12} \quad [8]$$

Solution

Let

$$I = \int_0^{\pi/2} \frac{\cos^5 \theta}{2 + \sin \theta} d\theta$$

Use the substitution

$$u = \sin \theta$$

so that

$$\frac{du}{d\theta} = \cos \theta \quad \Rightarrow \quad du = \cos \theta d\theta$$

Also,

$$\cos^5 \theta = \cos^4 \theta \cos \theta = (1 - \sin^2 \theta)^2 \cos \theta = (1 - u^2)^2 \cos \theta$$

When $\theta = 0$, $u = \sin 0 = 0$, and when $\theta = \frac{\pi}{2}$, $u = \sin \frac{\pi}{2} = 1$.

So the integral becomes

$$\begin{aligned} I &= \int_0^1 \frac{(1 - u^2)^2}{2 + u} du \\ &= \int_0^1 \frac{1 - 2u^2 + u^4}{u + 2} du \end{aligned}$$

Now divide $u^4 - 2u^2 + 1$ by $u + 2$:

$$\frac{u^4 - 2u^2 + 1}{u + 2} = u^3 - 2u^2 + 2u - 4 + \frac{9}{u + 2}$$

Hence

$$I = \int_0^1 \left(u^3 - 2u^2 + 2u - 4 + \frac{9}{u + 2} \right) du$$

Integrating term by term,

$$I = \left[\frac{u^4}{4} - \frac{2u^3}{3} + u^2 - 4u + 9 \ln(u + 2) \right]_0^1$$

Now substitute the limits:

$$\begin{aligned} I &= \left(\frac{1}{4} - \frac{2}{3} + 1 - 4 + 9 \ln 3 \right) - (0 - 0 + 0 - 0 + 9 \ln 2) \\ &= \frac{1}{4} - \frac{2}{3} + 1 - 4 + 9(\ln 3 - \ln 2) \\ &= \frac{3 - 8 + 12 - 48}{12} + 9 \ln\left(\frac{3}{2}\right) \\ &= -\frac{41}{12} + 9 \ln\left(\frac{3}{2}\right) \end{aligned}$$

Therefore,

$$\int_0^{\pi/2} \frac{\cos^5 \theta}{2 + \sin \theta} d\theta = 9 \ln\left(\frac{3}{2}\right) - \frac{41}{12}$$

as required.

2. (a) Use the substitution $u = 1 + x^2$ to show that

$$\int_0^2 \frac{x^9}{1+x^2} dx = \int_p^q \frac{(u-1)^4}{2u} du$$

where p and q are constants to be found.

[3]

- (b) Hence show that

$$\int_0^2 \frac{x^9}{1+x^2} dx = A + B \ln 5$$

where A and B are constants to be found.

[5]

Solution

- (a) Let

$$u = 1 + x^2$$

Then

$$\frac{du}{dx} = 2x \quad \Rightarrow \quad du = 2x dx$$

Also,

$$x^9 dx = x^8 (x dx) = (x^2)^4 \cdot \frac{du}{2} = (u-1)^4 \cdot \frac{du}{2}$$

and

$$1 + x^2 = u$$

So

$$\begin{aligned} \int_0^2 \frac{x^9}{1+x^2} dx &= \int \frac{x^9 dx}{1+x^2} \\ &= \int \frac{(u-1)^4/2}{u} du \end{aligned}$$

Now change the limits:

$$x = 0 \Rightarrow u = 1 + 0^2 = 1$$

$$x = 2 \Rightarrow u = 1 + 2^2 = 5$$

Hence

$$\int_0^2 \frac{x^9}{1+x^2} dx = \int_1^5 \frac{(u-1)^4}{2u} du$$

Therefore,

$$p = 1, \quad q = 5$$

- (b) From part (a),

$$\int_0^2 \frac{x^9}{1+x^2} dx = \int_1^5 \frac{(u-1)^4}{2u} du$$

Expand:

$$(u-1)^4 = u^4 - 4u^3 + 6u^2 - 4u + 1$$

$$\frac{(u-1)^4}{u} = u^3 - 4u^2 + 6u - 4 + \frac{1}{u}$$

So

$$\begin{aligned} \int_0^2 \frac{x^9}{1+x^2} dx &= \frac{1}{2} \int_1^5 \left(u^3 - 4u^2 + 6u - 4 + \frac{1}{u} \right) du \\ &= \left[\frac{u^4}{8} - \frac{2u^3}{3} + \frac{3u^2}{2} - 2u + \frac{1}{2} \ln u \right]_1^5 \end{aligned}$$

Now substitute the limits:

$$\begin{aligned}\int_0^2 \frac{x^9}{1+x^2} dx &= \left(\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 + \frac{1}{2} \ln 5 \right) - \left(\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2 + \frac{1}{2} \ln 1 \right) \\ &= \left(\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 + \frac{1}{2} \ln 5 \right) - \left(\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2 \right)\end{aligned}$$

Since

$$\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 = \frac{535}{24}$$

and

$$\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2 = -\frac{25}{24}$$

we get

$$\begin{aligned}\int_0^2 \frac{x^9}{1+x^2} dx &= \frac{535}{24} + \frac{1}{2} \ln 5 - \left(-\frac{25}{24} \right) \\ &= \frac{560}{24} + \frac{1}{2} \ln 5 \\ &= \frac{70}{3} + \frac{1}{2} \ln 5\end{aligned}$$

Hence

$$A = \frac{70}{3}, \quad B = \frac{1}{2}$$

3. Show that

$$\int \frac{\sqrt{x^2 - 16}}{x^2} dx = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2 - 16}}{x} + c \quad [6]$$

Solution

Let

$$I = \int \frac{\sqrt{x^2 - 16}}{x^2} dx$$

Use the substitution

$$x = 4 \cosh u$$

Then

$$\frac{dx}{du} = 4 \sinh u \quad \Rightarrow \quad dx = 4 \sinh u \, du$$

Also,

$$\begin{aligned} \sqrt{x^2 - 16} &= \sqrt{16 \cosh^2 u - 16} \\ &= 4\sqrt{\cosh^2 u - 1} \\ &= 4\sqrt{\sinh^2 u} \\ &= 4 \sinh u \end{aligned}$$

since $u > 0$ gives $\sinh u > 0$.
Substituting into the integral,

$$\begin{aligned} I &= \int \frac{4 \sinh u}{(4 \cosh u)^2} \cdot 4 \sinh u \, du \\ &= \int \frac{16 \sinh^2 u}{16 \cosh^2 u} \, du \\ &= \int \frac{\sinh^2 u}{\cosh^2 u} \, du \\ &= \int \tanh^2 u \, du \end{aligned}$$

Now use

$$\tanh^2 u = 1 - \operatorname{sech}^2 u$$

so

$$\begin{aligned} I &= \int (1 - \operatorname{sech}^2 u) \, du \\ &= u - \tanh u + c \end{aligned}$$

Now write this in terms of x .

From $x = 4 \cosh u$,

$$\cosh u = \frac{x}{4}$$

so, since $u > 0$,

$$u = \operatorname{arcosh}\left(\frac{x}{4}\right)$$

Also,

$$\begin{aligned} \tanh u &= \frac{\sinh u}{\cosh u} \\ &= \frac{\sqrt{x^2 - 16}/4}{x/4} \\ &= \frac{\sqrt{x^2 - 16}}{x} \end{aligned}$$

Therefore,

$$I = \operatorname{arcosh} \left(\frac{x}{4} \right) - \frac{\sqrt{x^2 - 16}}{x} + c$$

Hence,

$$\int \frac{\sqrt{x^2 - 16}}{x^2} dx = \operatorname{arcosh} \left(\frac{x}{4} \right) - \frac{\sqrt{x^2 - 16}}{x} + c$$

4. Find the exact value of

$$\int_0^{\frac{1}{2} \ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9 \tanh^2 x}} dx \quad [8]$$

Solution

Let

$$I = \int_0^{\frac{1}{2} \ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9 \tanh^2 x}} dx$$

Use the substitution

$$u = \frac{3}{2} \tanh x$$

Then

$$\frac{du}{dx} = \frac{3}{2} \operatorname{sech}^2 x$$

so

$$du = \frac{3}{2} \operatorname{sech}^2 x dx \quad \Rightarrow \quad \operatorname{sech}^2 x dx = \frac{2}{3} du$$

Also, from $u = \frac{3}{2} \tanh x$,

$$\tanh x = \frac{2u}{3}$$

so

$$\begin{aligned} \sqrt{4-9 \tanh^2 x} &= \sqrt{4-9 \left(\frac{2u}{3}\right)^2} \\ &= \sqrt{4-4u^2} \\ &= 2\sqrt{1-u^2} \end{aligned}$$

Now change the limits.

When $x = 0$,

$$u = \frac{3}{2} \tanh 0 = 0$$

When $x = \frac{1}{2} \ln(2 + \sqrt{3})$, we have

$$e^{2x} = 2 + \sqrt{3}$$

and

$$\begin{aligned} \tanh x &= \frac{e^{2x} - 1}{e^{2x} + 1} \\ &= \frac{(2 + \sqrt{3}) - 1}{(2 + \sqrt{3}) + 1} \\ &= \frac{1 + \sqrt{3}}{3 + \sqrt{3}} \end{aligned}$$

Rationalising,

$$\begin{aligned} \frac{1 + \sqrt{3}}{3 + \sqrt{3}} &= \frac{(1 + \sqrt{3})(3 - \sqrt{3})}{(3 + \sqrt{3})(3 - \sqrt{3})} \\ &= \frac{3 - \sqrt{3} + 3\sqrt{3} - 3}{9 - 3} \\ &= \frac{2\sqrt{3}}{6} \\ &= \frac{1}{\sqrt{3}} \end{aligned}$$

Hence

$$u = \frac{3}{2} \cdot \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{2}$$

So the integral becomes

$$\begin{aligned} I &= \int_0^{\sqrt{3}/2} \frac{\frac{2}{3} du}{2\sqrt{1-u^2}} \\ &= \frac{1}{3} \int_0^{\sqrt{3}/2} \frac{1}{\sqrt{1-u^2}} du \end{aligned}$$

Now

$$\int \frac{1}{\sqrt{1-u^2}} du = \arcsin u$$

so

$$\begin{aligned} I &= \frac{1}{3} [\arcsin u]_0^{\sqrt{3}/2} \\ &= \frac{1}{3} \left(\arcsin \frac{\sqrt{3}}{2} - \arcsin 0 \right) \\ &= \frac{1}{3} \left(\frac{\pi}{3} - 0 \right) \\ &= \frac{\pi}{9} \end{aligned}$$

Therefore,

$$\int_0^{\frac{1}{2} \ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9 \tanh^2 x}} dx = \frac{\pi}{9}$$

5. Using calculus, find the exact values of

(i)

$$\int_1^2 \frac{1}{x^2 + 3x + 2} dx \quad [3]$$

(ii)

$$\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2} dx \quad [5]$$

Solution

(i) First factorise the denominator:

$$x^2 + 3x + 2 = (x+1)(x+2)$$

Use partial fractions:

$$\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

So

$$1 = A(x+2) + B(x+1)$$

Putting $x = -1$ gives $A = 1$, and putting $x = -2$ gives $B = -1$. Hence

$$\frac{1}{x^2 + 3x + 2} = \frac{1}{x+1} - \frac{1}{x+2}$$

Therefore

$$\begin{aligned} \int_1^2 \frac{1}{x^2 + 3x + 2} dx &= \int_1^2 \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx \\ &= [\ln(x+1) - \ln(x+2)]_1^2 \\ &= (\ln 3 - \ln 4) - (\ln 2 - \ln 3) \\ &= 2 \ln 3 - \ln 4 - \ln 2 \\ &= \ln \left(\frac{9}{8} \right) \end{aligned}$$

(ii) Use the substitution

$$x = 2 \sin \theta$$

Then

$$\frac{dx}{d\theta} = 2 \cos \theta \quad \Rightarrow \quad dx = 2 \cos \theta d\theta$$

Also,

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2 \theta} = 2 \cos \theta$$

since on the interval we use, θ lies between $\frac{\pi}{6}$ and $\frac{\pi}{3}$, so $\cos \theta > 0$.

Now change the limits.

When $x = 1$,

$$1 = 2 \sin \theta \quad \Rightarrow \quad \sin \theta = \frac{1}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{6}$$

When $x = \sqrt{3}$,

$$\sqrt{3} = 2 \sin \theta \quad \Rightarrow \quad \sin \theta = \frac{\sqrt{3}}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{3}$$

So

$$\begin{aligned}\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2} dx &= \int_{\pi/6}^{\pi/3} \frac{2 \cos \theta}{(2 \sin \theta)^2} \cdot 2 \cos \theta d\theta \\ &= \int_{\pi/6}^{\pi/3} \frac{4 \cos^2 \theta}{4 \sin^2 \theta} d\theta \\ &= \int_{\pi/6}^{\pi/3} \cot^2 \theta d\theta\end{aligned}$$

Use the identity

$$\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$$

Then

$$\begin{aligned}\int_{\pi/6}^{\pi/3} \cot^2 \theta d\theta &= \int_{\pi/6}^{\pi/3} (\operatorname{cosec}^2 \theta - 1) d\theta \\ &= [-\cot \theta - \theta]_{\pi/6}^{\pi/3}\end{aligned}$$

Now evaluate:

$$\begin{aligned}[-\cot \theta - \theta]_{\pi/6}^{\pi/3} &= \left(-\cot \frac{\pi}{3} - \frac{\pi}{3}\right) - \left(-\cot \frac{\pi}{6} - \frac{\pi}{6}\right) \\ &= \left(-\frac{1}{\sqrt{3}} - \frac{\pi}{3}\right) - \left(-\sqrt{3} - \frac{\pi}{6}\right) \\ &= -\frac{\sqrt{3}}{3} - \frac{\pi}{3} + \sqrt{3} + \frac{\pi}{6} \\ &= \frac{2\sqrt{3}}{3} - \frac{\pi}{6}\end{aligned}$$

6. Find the exact value of

$$\int_2^4 \frac{\sqrt{x^2 - 4}}{x^2} dx$$

Give your answer in simplest exact form.

[9]

Solution

Let

$$I = \int_2^4 \frac{\sqrt{x^2 - 4}}{x^2} dx$$

Use the substitution

$$x = 2 \sec \theta$$

Then

$$\frac{dx}{d\theta} = 2 \sec \theta \tan \theta \quad \Rightarrow \quad dx = 2 \sec \theta \tan \theta d\theta$$

Also,

$$\begin{aligned} \sqrt{x^2 - 4} &= \sqrt{4 \sec^2 \theta - 4} \\ &= \sqrt{4(\sec^2 \theta - 1)} \\ &= \sqrt{4 \tan^2 \theta} \\ &= 2 \tan \theta \end{aligned}$$

For the limits:

When $x = 2$,

$$2 = 2 \sec \theta \Rightarrow \sec \theta = 1 \Rightarrow \theta = 0$$

When $x = 4$,

$$4 = 2 \sec \theta \Rightarrow \sec \theta = 2 \Rightarrow \theta = \frac{\pi}{3}$$

So the integral becomes

$$\begin{aligned} I &= \int_0^{\pi/3} \frac{2 \tan \theta}{(2 \sec \theta)^2} \cdot 2 \sec \theta \tan \theta d\theta \\ &= \int_0^{\pi/3} \frac{2 \tan \theta}{4 \sec^2 \theta} \cdot 2 \sec \theta \tan \theta d\theta \\ &= \int_0^{\pi/3} \frac{\tan^2 \theta}{\sec \theta} d\theta \end{aligned}$$

Now use $\tan^2 \theta = \sec^2 \theta - 1$:

$$\begin{aligned} \frac{\tan^2 \theta}{\sec \theta} &= \frac{\sec^2 \theta - 1}{\sec \theta} \\ &= \sec \theta - \frac{1}{\sec \theta} \\ &= \sec \theta - \cos \theta \end{aligned}$$

Hence

$$I = \int_0^{\pi/3} (\sec \theta - \cos \theta) d\theta$$

Integrating,

$$\begin{aligned} I &= [\ln(\sec \theta + \tan \theta) - \sin \theta]_0^{\pi/3} \\ &= \left(\ln(2 + \sqrt{3}) - \sin \frac{\pi}{3} \right) - (\ln(1 + 0) - \sin 0) \\ &= \ln(2 + \sqrt{3}) - \frac{\sqrt{3}}{2} \end{aligned}$$

Alternative method (using a hyperbolic substitution):

Use the substitution

$$x = 2 \cosh u$$

Then

$$\frac{dx}{du} = 2 \sinh u \quad \Rightarrow \quad dx = 2 \sinh u \, du$$

Also, since $u \geq 0$ on this interval,

$$\sqrt{x^2 - 4} = \sqrt{4 \cosh^2 u - 4} = 2 \sinh u$$

When $x = 2$, $\cosh u = 1$, so $u = 0$. When $x = 4$, $\cosh u = 2$, so $u = \operatorname{arcosh} 2 = \ln(2 + \sqrt{3})$.

So the integral becomes

$$\begin{aligned} I &= \int_0^{\ln(2+\sqrt{3})} \frac{2 \sinh u}{4 \cosh^2 u} \cdot 2 \sinh u \, du \\ &= \int_0^{\ln(2+\sqrt{3})} \tanh^2 u \, du \\ &= \int_0^{\ln(2+\sqrt{3})} (1 - \operatorname{sech}^2 u) \, du \\ &= [u - \tanh u]_0^{\ln(2+\sqrt{3})} \end{aligned}$$

At the upper limit, $\cosh u = 2$ and $\sinh u = \sqrt{\cosh^2 u - 1} = \sqrt{3}$, so $\tanh u = \frac{\sqrt{3}}{2}$. Hence

$$I = \ln(2 + \sqrt{3}) - \frac{\sqrt{3}}{2}$$

7. Show that

$$\int \frac{x^2}{\sqrt{x^2 + 16}} dx = \frac{1}{2}x\sqrt{x^2 + 16} - 8 \operatorname{arsinh}\left(\frac{x}{4}\right) + c \quad [6]$$

Solution

Let

$$x = 4 \sinh u$$

Then

$$\frac{dx}{du} = 4 \cosh u \quad \Rightarrow \quad dx = 4 \cosh u \, du$$

Also,

$$\begin{aligned} \sqrt{x^2 + 16} &= \sqrt{16 \sinh^2 u + 16} \\ &= \sqrt{16(\sinh^2 u + 1)} \\ &= \sqrt{16 \cosh^2 u} \\ &= 4 \cosh u \end{aligned}$$

since $\cosh u > 0$ for real u .

Now substitute into the integral:

$$\begin{aligned} \int \frac{x^2}{\sqrt{x^2 + 16}} dx &= \int \frac{(4 \sinh u)^2}{4 \cosh u} \cdot 4 \cosh u \, du \\ &= \int 16 \sinh^2 u \, du \\ &= 16 \int \sinh^2 u \, du \end{aligned}$$

Use the identity

$$\sinh^2 u = \frac{\cosh 2u - 1}{2}$$

So

$$\begin{aligned} 16 \int \sinh^2 u \, du &= 16 \int \frac{\cosh 2u - 1}{2} \, du \\ &= 8 \int (\cosh 2u - 1) \, du \\ &= 8 \left(\frac{1}{2} \sinh 2u - u \right) + c \\ &= 4 \sinh 2u - 8u + c \end{aligned}$$

Now use

$$\sinh 2u = 2 \sinh u \cosh u$$

giving

$$4 \sinh 2u - 8u = 8 \sinh u \cosh u - 8u$$

Now convert back to x .

From $x = 4 \sinh u$,

$$\sinh u = \frac{x}{4}$$

and since $\sqrt{x^2 + 16} = 4 \cosh u$,

$$\cosh u = \frac{\sqrt{x^2 + 16}}{4}$$

Also,

$$u = \operatorname{arsinh} \left(\frac{x}{4} \right)$$

Therefore

$$\begin{aligned} 8 \sinh u \cosh u - 8u + c &= 8 \left(\frac{x}{4} \right) \left(\frac{\sqrt{x^2 + 16}}{4} \right) - 8 \operatorname{arsinh} \left(\frac{x}{4} \right) + c \\ &= \frac{1}{2} x \sqrt{x^2 + 16} - 8 \operatorname{arsinh} \left(\frac{x}{4} \right) + c \end{aligned}$$

Hence

$$\int \frac{x^2}{\sqrt{x^2 + 16}} \, dx = \frac{1}{2} x \sqrt{x^2 + 16} - 8 \operatorname{arsinh} \left(\frac{x}{4} \right) + c$$

8. Show that

$$\int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}} dx = \frac{32\sqrt{6} - 30\sqrt{5}}{5} \quad [7]$$

Solution

Let

$$I = \int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}} dx$$

Use the substitution $u = e^x$. Then

$$\frac{du}{dx} = e^x = u \quad \Rightarrow \quad du = e^x dx, \quad dx = \frac{du}{u}$$

Also, the limits change as follows:

$$x = 0 \Rightarrow u = 1, \quad x = \ln 2 \Rightarrow u = 2$$

So

$$\begin{aligned} I &= \int_1^2 \frac{u^3 + 2u^2}{\sqrt{u + 4}} \cdot \frac{1}{u} du \\ &= \int_1^2 \frac{u^2 + 2u}{\sqrt{u + 4}} du \end{aligned}$$

Now use a second substitution:

$$v = u + 4 \quad \Rightarrow \quad dv = du, \quad u = v - 4$$

The new limits are

$$u = 1 \Rightarrow v = 5, \quad u = 2 \Rightarrow v = 6$$

Therefore

$$\begin{aligned} I &= \int_5^6 \frac{(v - 4)^2 + 2(v - 4)}{\sqrt{v}} dv \\ &= \int_5^6 \frac{v^2 - 8v + 16 + 2v - 8}{\sqrt{v}} dv \\ &= \int_5^6 \frac{v^2 - 6v + 8}{\sqrt{v}} dv \\ &= \int_5^6 \left(v^{3/2} - 6v^{1/2} + 8v^{-1/2} \right) dv \end{aligned}$$

Integrate term by term:

$$I = \left[\frac{2}{5} v^{5/2} - 4v^{3/2} + 16v^{1/2} \right]_5^6$$

Now substitute the limits:

$$\begin{aligned} I &= \left(\frac{2}{5} \cdot 6^{5/2} - 4 \cdot 6^{3/2} + 16 \cdot 6^{1/2} \right) - \left(\frac{2}{5} \cdot 5^{5/2} - 4 \cdot 5^{3/2} + 16 \cdot 5^{1/2} \right) \\ &= \left(\frac{2}{5} \cdot 36\sqrt{6} - 4 \cdot 6\sqrt{6} + 16\sqrt{6} \right) - \left(\frac{2}{5} \cdot 25\sqrt{5} - 4 \cdot 5\sqrt{5} + 16\sqrt{5} \right) \\ &= \left(\frac{72}{5}\sqrt{6} - 24\sqrt{6} + 16\sqrt{6} \right) - \left(10\sqrt{5} - 20\sqrt{5} + 16\sqrt{5} \right) \\ &= \left(\frac{72}{5} - 8 \right) \sqrt{6} - 6\sqrt{5} \\ &= \frac{32}{5} \sqrt{6} - 6\sqrt{5} \\ &= \frac{32\sqrt{6} - 30\sqrt{5}}{5} \end{aligned}$$

Hence

$$\int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}} \, dx = \frac{32\sqrt{6} - 30\sqrt{5}}{5}$$

as required.

9. Using the substitution $u = \frac{2}{3} \tanh x$, evaluate

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4 \tanh^2 x}} dx \quad [8]$$

Solution

Let

$$I = \int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4 \tanh^2 x}} dx$$

Use the given substitution

$$u = \frac{2}{3} \tanh x$$

Differentiate:

$$\frac{du}{dx} = \frac{2}{3} \operatorname{sech}^2 x$$

so

$$du = \frac{2}{3} \operatorname{sech}^2 x dx$$

and hence

$$\operatorname{sech}^2 x dx = \frac{3}{2} du$$

Also, from $u = \frac{2}{3} \tanh x$,

$$\tanh x = \frac{3}{2} u$$

so the expression inside the square root becomes

$$\begin{aligned} 9 - 4 \tanh^2 x &= 9 - 4 \left(\frac{3}{2} u \right)^2 \\ &= 9 - 4 \cdot \frac{9}{4} u^2 \\ &= 9 - 9u^2 \\ &= 9(1 - u^2) \end{aligned}$$

Therefore,

$$\sqrt{9 - 4 \tanh^2 x} = 3\sqrt{1 - u^2}$$

Now change the limits.

When $x = 0$,

$$u = \frac{2}{3} \tanh 0 = 0$$

When $x = 1$,

$$u = \frac{2}{3} \tanh 1$$

Substituting everything into the integral:

$$\begin{aligned} I &= \int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4 \tanh^2 x}} dx \\ &= \int_0^{\frac{2}{3} \tanh 1} \frac{\frac{3}{2} du}{3\sqrt{1 - u^2}} \\ &= \frac{1}{2} \int_0^{\frac{2}{3} \tanh 1} \frac{1}{\sqrt{1 - u^2}} du \end{aligned}$$

Using

$$\int \frac{1}{\sqrt{1 - u^2}} du = \arcsin u$$

we get

$$\begin{aligned} I &= \frac{1}{2} [\arcsin u]_0^{\frac{2}{3} \tanh 1} \\ &= \frac{1}{2} \arcsin\left(\frac{2}{3} \tanh 1\right) - \frac{1}{2} \arcsin(0) \\ &= \frac{1}{2} \arcsin\left(\frac{2}{3} \tanh 1\right) \end{aligned}$$

So the exact value is

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4 \tanh^2 x}} \, dx = \frac{1}{2} \arcsin\left(\frac{2}{3} \tanh 1\right)$$

Numerically,

$$\frac{1}{2} \arcsin\left(\frac{2}{3} \tanh 1\right) \approx 0.266$$

10. Show that

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2 + \cos \theta} d\theta = 15 \ln \left(\frac{3}{2} \right) - 6 \quad [7]$$

Solution

Let

$$I = \int_0^{\pi/2} \frac{\sin 3\theta}{2 + \cos \theta} d\theta$$

First rewrite $\sin 3\theta$ in terms of $\sin \theta$ and $\cos \theta$.

Using

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

and $\sin^2 \theta = 1 - \cos^2 \theta$,

$$\begin{aligned} \sin 3\theta &= \sin \theta (3 - 4 \sin^2 \theta) \\ &= \sin \theta (3 - 4(1 - \cos^2 \theta)) \\ &= \sin \theta (4 \cos^2 \theta - 1) \end{aligned}$$

So

$$I = \int_0^{\pi/2} \frac{\sin \theta (4 \cos^2 \theta - 1)}{2 + \cos \theta} d\theta$$

Now use the substitution

$$u = \cos \theta \quad \Rightarrow \quad du = -\sin \theta d\theta$$

When $\theta = 0$, $u = 1$.

When $\theta = \frac{\pi}{2}$, $u = 0$.

Hence

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\sin \theta (4 \cos^2 \theta - 1)}{2 + \cos \theta} d\theta \\ &= \int_1^0 \frac{4u^2 - 1}{u + 2} (-du) \\ &= \int_0^1 \frac{4u^2 - 1}{u + 2} du \end{aligned}$$

Now divide $4u^2 - 1$ by $u + 2$:

$$4u^2 - 1 = (u + 2)(4u - 8) + 15$$

So

$$\frac{4u^2 - 1}{u + 2} = 4u - 8 + \frac{15}{u + 2}$$

Therefore

$$\begin{aligned} I &= \int_0^1 \left(4u - 8 + \frac{15}{u + 2} \right) du \\ &= [2u^2 - 8u + 15 \ln(u + 2)]_0^1 \end{aligned}$$

Now substitute the limits:

$$\begin{aligned} I &= (2(1)^2 - 8(1) + 15 \ln 3) - (2(0)^2 - 8(0) + 15 \ln 2) \\ &= (2 - 8 + 15 \ln 3) - 15 \ln 2 \\ &= -6 + 15 \ln \left(\frac{3}{2} \right) \end{aligned}$$

Hence

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2 + \cos \theta} d\theta = 15 \ln \left(\frac{3}{2} \right) - 6$$

So the required result is shown.

11. (a) Use the substitution $u = \sqrt{2x+1}$ to show that

$$\int_0^{15/2} \frac{x}{2 + \sqrt{2x+1}} dx = \int_p^q \frac{u(u^2-1)}{2(u+2)} du$$

where p and q are constants to be found.

[3]

- (b) Hence show that

$$\int_0^{15/2} \frac{x}{2 + \sqrt{2x+1}} dx = A - B \ln 2$$

where A and B are constants to be found.

[4]

Solution

- (a) Let

$$u = \sqrt{2x+1}$$

Then

$$u^2 = 2x+1 \quad \Rightarrow \quad x = \frac{u^2-1}{2}$$

Differentiate $u^2 = 2x+1$ with respect to x :

$$2u \frac{du}{dx} = 2$$

so

$$\frac{dx}{du} = u \quad \Rightarrow \quad dx = u du$$

Also,

$$\sqrt{2x+1} = u$$

Now change the limits.

When $x = 0$,

$$u = \sqrt{2(0)+1} = 1$$

When $x = \frac{15}{2}$,

$$u = \sqrt{2 \cdot \frac{15}{2} + 1} = \sqrt{16} = 4$$

Therefore

$$\begin{aligned} \int_0^{15/2} \frac{x}{2 + \sqrt{2x+1}} dx &= \int_1^4 \frac{\frac{u^2-1}{2}}{2+u} \cdot u du \\ &= \int_1^4 \frac{u(u^2-1)}{2(u+2)} du \end{aligned}$$

Hence $p = 1$ and $q = 4$.

- (b) Using part (a),

$$\int_0^{15/2} \frac{x}{2 + \sqrt{2x+1}} dx = \int_1^4 \frac{u(u^2-1)}{2(u+2)} du$$

First simplify the integrand. Since

$$u(u^2-1) = u^3 - u$$

divide $u^3 - u$ by $u+2$:

$$u^3 - u = (u+2)(u^2 - 2u + 3) - 6$$

So

$$\frac{u(u^2 - 1)}{2(u + 2)} = \frac{1}{2} \left(u^2 - 2u + 3 - \frac{6}{u + 2} \right)$$

Hence

$$\begin{aligned} \int_1^4 \frac{u(u^2 - 1)}{2(u + 2)} \, du &= \frac{1}{2} \int_1^4 \left(u^2 - 2u + 3 - \frac{6}{u + 2} \right) \, du \\ &= \frac{1}{2} \left[\frac{u^3}{3} - u^2 + 3u - 6 \ln(u + 2) \right]_1^4 \end{aligned}$$

Now substitute the limits:

$$\begin{aligned} &= \frac{1}{2} \left[\left(\frac{4^3}{3} - 4^2 + 3 \cdot 4 - 6 \ln 6 \right) - \left(\frac{1^3}{3} - 1^2 + 3 \cdot 1 - 6 \ln 3 \right) \right] \\ &= \frac{1}{2} \left[\left(\frac{64}{3} - 16 + 12 - 6 \ln 6 \right) - \left(\frac{1}{3} - 1 + 3 - 6 \ln 3 \right) \right] \\ &= \frac{1}{2} \left[\left(\frac{52}{3} - 6 \ln 6 \right) - \left(\frac{7}{3} - 6 \ln 3 \right) \right] \\ &= \frac{1}{2} \left[\frac{45}{3} - 6(\ln 6 - \ln 3) \right] \\ &= \frac{1}{2} \left[15 - 6 \ln \left(\frac{6}{3} \right) \right] \\ &= \frac{1}{2} (15 - 6 \ln 2) \\ &= \frac{15}{2} - 3 \ln 2 \end{aligned}$$

Therefore

$$\int_0^{15/2} \frac{x}{2 + \sqrt{2x + 1}} \, dx = \frac{15}{2} - 3 \ln 2$$

So

$$A = \frac{15}{2}, \quad B = 3$$

12. Use an appropriate substitution to find the exact value of

$$\int_0^{\pi/6} \frac{\cos x \cos 2x}{(1 + \sin x)^{3/2}} dx$$

Give your answer in the form $a + b\sqrt{6}$, where a and b are rational numbers to be determined.

[7]

Solution

Let

$$I = \int_0^{\pi/6} \frac{\cos x \cos 2x}{(1 + \sin x)^{3/2}} dx$$

Use the substitution

$$u = 1 + \sin x$$

Then

$$\frac{du}{dx} = \cos x \quad \Rightarrow \quad du = \cos x dx$$

The limits change as follows:

$$\begin{aligned} x = 0 &\Rightarrow u = 1 + \sin 0 = 1 \\ x = \frac{\pi}{6} &\Rightarrow u = 1 + \sin \frac{\pi}{6} = 1 + \frac{1}{2} = \frac{3}{2} \end{aligned}$$

Also,

$$\sin x = u - 1$$

so

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2(u - 1)^2$$

Now expand:

$$\begin{aligned} \cos 2x &= 1 - 2(u^2 - 2u + 1) \\ &= 1 - 2u^2 + 4u - 2 \\ &= -2u^2 + 4u - 1 \end{aligned}$$

So the integral becomes

$$\begin{aligned} I &= \int_1^{3/2} \frac{\cos 2x}{u^{3/2}} du \\ &= \int_1^{3/2} \frac{-2u^2 + 4u - 1}{u^{3/2}} du \\ &= \int_1^{3/2} \left(-2u^{1/2} + 4u^{-1/2} - u^{-3/2} \right) du \end{aligned}$$

Integrating term by term,

$$\begin{aligned} I &= \left[-2 \cdot \frac{2}{3} u^{3/2} + 4 \cdot 2u^{1/2} - \frac{u^{-1/2}}{-1/2} \right]_1^{3/2} \\ &= \left[-\frac{4}{3} u^{3/2} + 8u^{1/2} + 2u^{-1/2} \right]_1^{3/2} \end{aligned}$$

Now evaluate at $u = \frac{3}{2}$:

$$\begin{aligned} \left(\frac{3}{2} \right)^{1/2} &= \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2} \\ \left(\frac{3}{2} \right)^{3/2} &= \frac{3}{2} \cdot \frac{\sqrt{6}}{2} = \frac{3\sqrt{6}}{4} \end{aligned}$$

$$\left(\frac{3}{2}\right)^{-1/2} = \frac{1}{\sqrt{3/2}} = \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$$

Hence

$$\begin{aligned} -\frac{4}{3} \left(\frac{3\sqrt{6}}{4} \right) + 8 \left(\frac{\sqrt{6}}{2} \right) + 2 \left(\frac{\sqrt{6}}{3} \right) &= -\sqrt{6} + 4\sqrt{6} + \frac{2\sqrt{6}}{3} \\ &= \frac{11\sqrt{6}}{3} \end{aligned}$$

At $u = 1$,

$$\begin{aligned} -\frac{4}{3}(1)^{3/2} + 8(1)^{1/2} + 2(1)^{-1/2} &= -\frac{4}{3} + 8 + 2 \\ &= \frac{26}{3} \end{aligned}$$

Therefore

$$\begin{aligned} I &= \frac{11\sqrt{6}}{3} - \frac{26}{3} \\ &= \frac{11\sqrt{6} - 26}{3} \end{aligned}$$

So the exact value is

$$-\frac{26}{3} + \frac{11}{3}\sqrt{6}$$

Hence

$$a = -\frac{26}{3}, \quad b = \frac{11}{3}$$

13. Using calculus, find the exact values of

(i)

$$\int_{-2}^0 \frac{1}{x^2 + 4x + 8} dx \quad [3]$$

(ii)

$$\int_0^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx \quad [5]$$

Solution

(i) Complete the square in the denominator:

$$x^2 + 4x + 8 = (x + 2)^2 + 4$$

Use the standard result

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + c$$

with $a = 2$, and with x replaced by $x + 2$:

$$\begin{aligned} \int_{-2}^0 \frac{1}{x^2 + 4x + 8} dx &= \int_{-2}^0 \frac{1}{(x + 2)^2 + 4} dx \\ &= \left[\frac{1}{2} \arctan\left(\frac{x + 2}{2}\right) \right]_{-2}^0 \\ &= \frac{1}{2} (\arctan 1 - \arctan 0) \\ &= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) \\ &= \frac{\pi}{8} \end{aligned}$$

(ii) Use the substitution

$$x = 2 \sin \theta$$

Then

$$dx = 2 \cos \theta d\theta$$

and

$$\sqrt{4 - x^2} = \sqrt{4 - 4 \sin^2 \theta} = 2 \cos \theta$$

Since $0 \leq x \leq \sqrt{3}$, we have $0 \leq \theta \leq \frac{\pi}{3}$, so $\cos \theta > 0$.

Change the limits:

$$x = 0 \Rightarrow \theta = 0, \quad x = \sqrt{3} \Rightarrow 2 \sin \theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$$

Now substitute:

$$\begin{aligned} \int_0^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx &= \int_0^{\pi/3} \frac{(2 \sin \theta)^2}{2 \cos \theta} \cdot 2 \cos \theta d\theta \\ &= \int_0^{\pi/3} 4 \sin^2 \theta d\theta \end{aligned}$$

Use

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

So

$$\begin{aligned}\int_0^{\pi/3} 4 \sin^2 \theta \, d\theta &= 4 \int_0^{\pi/3} \frac{1 - \cos 2\theta}{2} \, d\theta \\ &= 2 \int_0^{\pi/3} (1 - \cos 2\theta) \, d\theta \\ &= 2 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/3}\end{aligned}$$

Now evaluate:

$$\begin{aligned}2 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/3} &= 2 \left(\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} \right) \\ &= 2 \left(\frac{\pi}{3} - \frac{1}{2} \cdot \frac{\sqrt{3}}{2} \right) \\ &= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}\end{aligned}$$

14. Show that

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}} dx = \frac{16}{15}(7-4\sqrt{2}) \quad [7]$$

Solution

Let

$$I = \int_1^3 \frac{(x-1)^2}{\sqrt{x+1}} dx$$

Use the substitution

$$u = x + 1$$

so that

$$x = u - 1, \quad x - 1 = u - 2, \quad dx = du$$

The limits also change:

$$x = 1 \Rightarrow u = 2, \quad x = 3 \Rightarrow u = 4$$

So

$$\begin{aligned} I &= \int_2^4 \frac{(u-2)^2}{\sqrt{u}} du \\ &= \int_2^4 (u-2)^2 u^{-1/2} du \\ &= \int_2^4 (u^2 - 4u + 4) u^{-1/2} du \\ &= \int_2^4 (u^{3/2} - 4u^{1/2} + 4u^{-1/2}) du \end{aligned}$$

Integrating term by term,

$$\begin{aligned} I &= \left[\frac{2}{5} u^{5/2} - 4 \cdot \frac{2}{3} u^{3/2} + 4 \cdot 2u^{1/2} \right]_2^4 \\ &= \left[\frac{2}{5} u^{5/2} - \frac{8}{3} u^{3/2} + 8u^{1/2} \right]_2^4 \end{aligned}$$

Now evaluate at the limits.

When $u = 4$,

$$4^{1/2} = 2, \quad 4^{3/2} = 8, \quad 4^{5/2} = 32$$

so

$$\frac{2}{5}(32) - \frac{8}{3}(8) + 8(2) = \frac{64}{5} - \frac{64}{3} + 16$$

When $u = 2$,

$$2^{1/2} = \sqrt{2}, \quad 2^{3/2} = 2\sqrt{2}, \quad 2^{5/2} = 4\sqrt{2}$$

so

$$\frac{2}{5}(4\sqrt{2}) - \frac{8}{3}(2\sqrt{2}) + 8\sqrt{2} = \frac{8\sqrt{2}}{5} - \frac{16\sqrt{2}}{3} + 8\sqrt{2}$$

Hence

$$\begin{aligned} I &= \left(\frac{64}{5} - \frac{64}{3} + 16 \right) - \left(\frac{8\sqrt{2}}{5} - \frac{16\sqrt{2}}{3} + 8\sqrt{2} \right) \\ &= \frac{192 - 320 + 240}{15} - \frac{24\sqrt{2} - 80\sqrt{2} + 120\sqrt{2}}{15} \\ &= \frac{112}{15} - \frac{64\sqrt{2}}{15} \\ &= \frac{16}{15}(7 - 4\sqrt{2}) \end{aligned}$$

Therefore

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}} dx = \frac{16}{15}(7-4\sqrt{2})$$

as required.

15. Use an appropriate substitution to find the exact value of

$$\int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}} dx$$

Give your answer in the form $a + b\pi$, where a and b are rational numbers.

[9]

Solution

Use the substitution

$$x = 4 \sin \theta$$

Then

$$\frac{dx}{d\theta} = 4 \cos \theta \quad \Rightarrow \quad dx = 4 \cos \theta d\theta$$

Also,

$$\begin{aligned} \sqrt{16-x^2} &= \sqrt{16-16\sin^2 \theta} \\ &= \sqrt{16(1-\sin^2 \theta)} \\ &= \sqrt{16\cos^2 \theta} \\ &= 4 \cos \theta \end{aligned}$$

Since the limits will give $0 \leq \theta \leq \frac{\pi}{4}$, we have $\cos \theta \geq 0$, so this square root is indeed $4 \cos \theta$.
Now change the limits.

When $x = 0$,

$$0 = 4 \sin \theta \quad \Rightarrow \quad \sin \theta = 0 \quad \Rightarrow \quad \theta = 0$$

When $x = 2\sqrt{2}$,

$$2\sqrt{2} = 4 \sin \theta \quad \Rightarrow \quad \sin \theta = \frac{\sqrt{2}}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{4}$$

So the integral becomes

$$\begin{aligned} \int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}} dx &= \int_0^{\pi/4} \frac{(4 \sin \theta)^2}{4 \cos \theta} \cdot 4 \cos \theta d\theta \\ &= \int_0^{\pi/4} \frac{16 \sin^2 \theta}{4 \cos \theta} \cdot 4 \cos \theta d\theta \\ &= 16 \int_0^{\pi/4} \sin^2 \theta d\theta \end{aligned}$$

Use the identity

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

Then

$$\begin{aligned} 16 \int_0^{\pi/4} \sin^2 \theta d\theta &= 16 \int_0^{\pi/4} \frac{1 - \cos 2\theta}{2} d\theta \\ &= 8 \int_0^{\pi/4} (1 - \cos 2\theta) d\theta \\ &= 8 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/4} \end{aligned}$$

Now substitute the limits:

$$\begin{aligned} 8 \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/4} &= 8 \left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right) - 8 \left(0 - \frac{1}{2} \sin 0 \right) \\ &= 8 \left(\frac{\pi}{4} - \frac{1}{2} \right) \\ &= 2\pi - 4 \end{aligned}$$

So in the form $a + b\pi$,

$$a = -4, \quad b = 2$$